

(54) Title of the invention : A MEDICATION DISPENSING APPARATUS

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(57) Abstract :

The present invention is related to a medication dispensing apparatus is disclosed, designed for healthcare applications to enhance patient adherence and ensure accurate medication delivery. The apparatus integrates a servo-driven rotating tray (104) for automated pill dispensing, a liquid medicine dispenser (103) with LED indicators (112) for guided manual liquid administration, and a privacy-protected camera module (111) with a privacy cap (101) actuated by a servo motor (108). A touch sensor (110) detects pill pickup, triggering the camera module (111) to capture ingestion events, which are verified in real-time by a locally executed AI model on a microcontroller unit (106). The system includes a buzzer (109) that remains active until ingestion is confirmed and locally stores dispensing and verification data for offline operation. The apparatus operates without internet connectivity, ensuring privacy and reliability. Principal applications include clinical and home-care settings, addressing challenges in medication adherence and dosage accuracy for both pills and liquid medications.

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